

VEMULA MANOJ KUMAR

Network Engineer

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PROFESSIONAL SUMMARY

CCNA-trained Network Engineer with hands-on experience configuring, managing, and troubleshooting enterprise-level networks using Cisco routers and switches. Skilled in routing protocols, switching technologies, and network services with practical exposure to real-time lab environments using GNS3 and Cisco Packet Tracer. Skilled in packet-level analysis, network automation basics and Linux-based networking.

TECHNICAL SKILLS

Routing & Switching: OSPF (Multi-Area), Static & Default Routing, Basic BGP, VLANs, STP, Inter-VLAN Routing

Network Services: DHCP, DNS, NAT, ARP, TCP/IP, OSI Model, IP Addressing & Subnetting

Cisco Hardware: 1900 Series Routers, Catalyst 2960 Switches, Layer 3 3850 Switches

Tools & Simulators: GNS3, Cisco Packet Tracer, Wireshark

CLI Diagnostics: ping, traceroute, show ip route, show ip ospf neighbour, show vlan brief, debug commands

Linux Networking: IP configuration, SSH setup, basic firewall configuration (iptables/firewall), network troubleshooting (netstat, tcpdump)

HANDS-ON HARDWARE EXPERIENCE

Cisco Real-Device Lab Practice

- Configured and managed Cisco 1900 Series routers, Catalyst 2960 switches, and Layer 3 3850 switches on live production-grade hardware
- Implemented OSPF, VLANs, SVIs, and Inter-VLAN routing on Layer 3 switches; verified traffic segmentation and routing across all VLANs
- Implemented EtherChannel (LACP/PAgP) for link aggregation and redundancy between switches
- Configured HSRP and VRRP for first-hop gateway redundancy; tested failover behaviour and virtual IP assignment
- Configured DHCP server and DHCP Relay Agent (ip helper-address) to serve clients across multiple subnets
- Configured standard and extended ACLs to control inter-VLAN traffic and restrict unauthorized access.
- Diagnosed and resolved ARP, ICMP, and routing failures using show, debug, and packet capture techniques

NETWORKING PROJECTS

Enterprise Multi-Area OSPF Deployment

GNS3

- Designed and implemented multi-area OSPF topology including Backbone, Stub, and NSSA areas
- Configured route summarization and optimized LSA propagation across Types 1–7
- Resolved OSPF adjacency failures caused by timer mismatch & incorrect network types
- Ensured fast convergence and high availability; verified neighbour relationships using Cisco CLI

Static & Default Routing – Branch Network Design

- Designed branch network topology with full subnetting, VLANs, and router role assignment
- Configured static and default routes for reliable communication between branch and central network
- Verified routing paths and documented configurations for operational handover

DHCP Network Implementation & Optimization

GNS3 + Packet Tracer

- Configured DHCP servers for automated IP allocation across multiple subnets
- Analysed full DORA packet exchange sequence for protocol-level troubleshooting understanding
- Reduced IP conflicts by optimizing DHCP scope and lease configurations
- Minimized IP conflicts and optimized DHCP scope settings for efficient address management

ARP & ICMP Packet Analysis & Troubleshooting

- Captured and analysed ARP and ICMP traffic to diagnose packet behaviour across subnets
- Diagnosed inter-VLAN communication failure due to incorrect trunk configuration and resolved issue
- Used ping and traceroute to identify and resolve connectivity and delivery issues

Network Reconnaissance & Automation Tool (Python)

- Developed a Python-based tool for automated network asset and domain discovery
- Performed subdomain enumeration and DNS resolution to identify active endpoints
- Generated structured outputs for improved network visibility and analysis

INTERNSHIP

Network Engineer Intern – Tungabhadra Networks

Network Configuration & Troubleshooting | Duration: [Add dates]

- Assisted in configuring VLANs, routing and basic network services on Cisco devices
- Performed troubleshooting tasks on routing and switching issues for live infrastructure
- Supported network maintenance activities in a production-like environment

CERTIFICATIONS & TRAINING

- CCNA – Cisco Certified Network Associate (In Progress)
- Internship Certification – Tungabhadra Networks (Network Configuration & Troubleshooting)
- TryHackMe – Hands-on labs: networking, penetration testing, web vulnerabilities

EDUCATION

B.Tech – Electronics and Communication Engineering

2021 – May 2025

CMR Institute of Technology, Medchal

Intermediate (MPC)

Alphores Junior College, Karimnagar

SSC

Siddhartha High School, Karimnagar

LANGUAGES

English | Telugu | Hindi

I hereby declare that the information furnished above is true to the best of my knowledge.

Vemula Manoj Kumar